Pressure, Temperature, Humidity, Visibility and Particulate

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This chapter introduces a set of atmospheric quantities that are used repeatedly in environmental models and in the associated practical work. The aim is to connect the definitions to measurements and to simple calculations: pressure and temperature describe the thermodynamic state of the air; humidity controls when water can condense; aerosol particles interact with water vapour; and particle size and number affect both visibility and particulate-matter loading.

By the end of the chapter you should be able to:

- use the ideal-gas law in molar and density forms;
- distinguish environmental lapse rate from the cooling rate of rising air parcel;
- calculate water-vapour pressure, saturation vapour pressure, relative humidity and water-vapour mixing ratio;
- explain why soluble aerosol particles grow as relative humidity increases;
- relate particle size and number conc. to light extinction and visual range;
- calculate simplified particulate-matter mass concentrations.

1 Pressure

For a gas that behaves approximately ideally,

$$PV = nR_{\text{gas}}T, \tag{1}$$

where P is pressure, V is volume, n is the amount of gas in moles, $R_{\text{gas}} = 8.314$ J mol^{-1} K^{-1} is the universal gas constant and T is absolute temperature.

For dry air it is often more useful to write the ideal-gas law in terms of density. Defining the specific gas constant for dry air as

$$R_a = \frac{R_{\text{gas}}}{M_a} \simeq 287 \text{ J kg}^{-1}\text{ K}^{-1}, \tag{2}$$

where $M_a \simeq 28.97 \times 10^{-3}$ kg mol^{-1} is the molar mass of dry air, gives

$$P = \rho_a R_a T. \tag{3}$$

Thus, at a fixed temperature, higher pressure corresponds to greater air density; at a fixed density, increasing temperature increases pressure.

Near the surface, atmospheric pressure is about 10^5 Pa. A pressure of 10^5 Pa means a force of about 10^5 N acts on each square metre of a horizontal surface. This is the consequence of the weight of the atmosphere above us, transmitted through the pressure field.

2 Temperature

Temperature is related to the microscopic kinetic energy of molecules. In an environmental model, however, the important point is usually how temperature changes when energy is transferred or when air expands and contracts.

Air temperature can change through several processes, including:

- sensible-heat transfer between the surface and the atmosphere;
- convection and advection, which transport heat with moving air;
- absorption and emission of radiation by gases, clouds and particles;
- phase changes of water, which exchange latent heat;
- chemical or other transformations that release or absorb energy.

A particularly important atmospheric process is *adiabatic expansion*. As an air parcel rises, the surrounding pressure decreases. The parcel expands and does work on its surroundings. If heat exchange with the environment is small during the ascent, its internal energy and temperature decrease. A descending parcel is compressed and warms for the same reason.

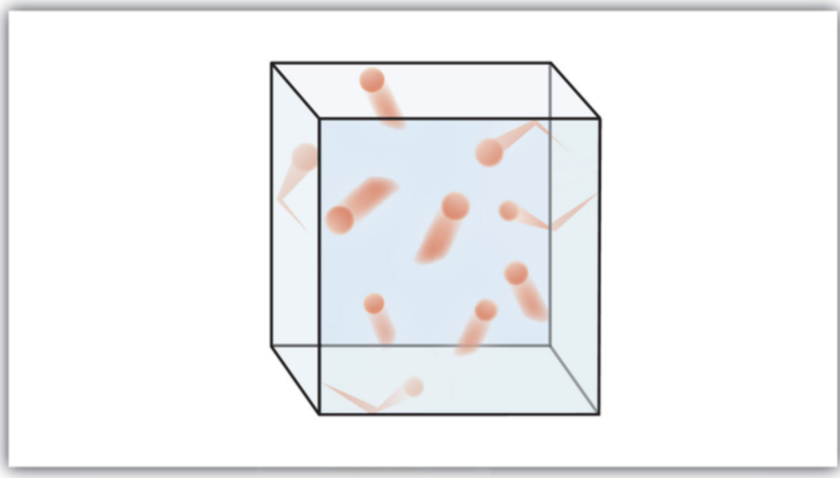

Figure 1: A gas consists of molecules in continual random motion. Collisions with the walls of a container exert pressure; increasing molecular kinetic energy corresponds to a higher temperature. Source attributions: PEOI introductory chemistry materials.

3 Parcel lapse rate and environmental lapse rate

A *lapse rate* describes how temperature changes with height. Two different ideas should not be confused:

- the **environmental lapse rate** is the temperature profile of the surrounding atmosphere at a particular place and time;
- a **parcel lapse rate** describes how the temperature of a moving air parcel changes as it rises or sinks.

For an unsaturated parcel rising adiabatically, the dry adiabatic lapse rate is approximately 9.8 K km^{-1}; in the exercises below we round this to 10 K km^{-1}. Once a rising parcel becomes saturated, condensation releases latent heat and reduces the rate of cooling. The saturated (moist) adiabatic lapse rate is not a single constant—it depends on temperature and moisture—but a representative value of about 6 K km^{-1} is used in these exercises.

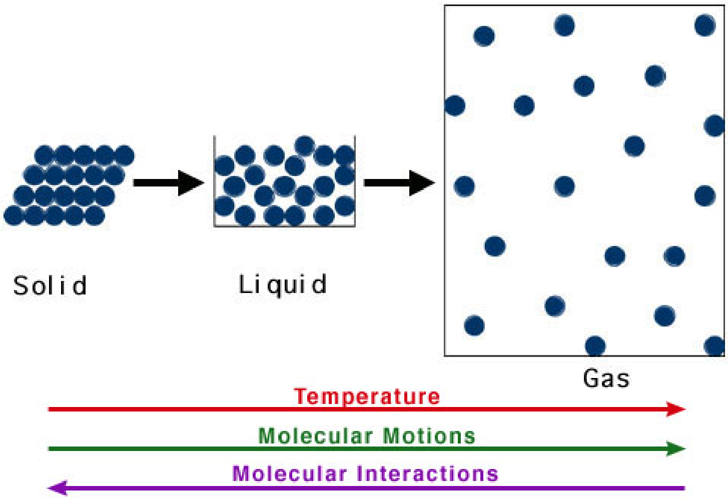

Figure 2: Molecular motion increases from solid to liquid to gas. During condensation, water molecules move from the gas phase to the liquid phase and latent heat is released to the surrounding air. Source attribution: Washington University thermochemistry tutorial.

The comparison between the parcel temperature and the environmental temperature is central to atmospheric stability. If a displaced parcel remains warmer and therefore less dense than its environment, it tends to continue rising; if it becomes colder and denser than its environment, its ascent tends to be suppressed.

When water vapour condenses, latent heat is released. Around typical atmospheric temperatures the latent heat of vaporisation is approximately

$$L_v \simeq 2.5 \times 10^6 \ \mathrm{J\,kg^{-1}}. \tag{4}$$

4 Humidity, partial pressure and mixing ratio

The atmosphere is a mixture of gases. Each gas contributes a *partial pressure*; the total pressure is the sum of the partial pressures. We use e for the partial pressure of water vapour. Water vapour approximately obeys its own ideal-gas law,

$$e = \rho_v R_v T, \tag{5}$$

where ρ_v is the mass of water vapour per unit volume and $R_v \simeq 461$ J kg^{-1} K^{-1} is the specific gas constant for water vapour.

4.1 Saturation vapour pressure and relative humidity

At a given temperature there is an equilibrium vapour pressure over a flat surface of liquid water, called the *saturation vapour pressure*, $e_s(T)$. It increases rapidly with temperature. A convenient integrated form of the Clausius–Clapeyron relation for the temperature range used here is

$$e_s(T) = 610.7 \exp\left[\frac{L_v}{R_v}\left(\frac{1}{273.15} - \frac{1}{T}\right)\right]. \tag{6}$$

Relative humidity is then

$$RH = \frac{e}{e_s(T)}. \tag{7}$$

A value $RH = 1$ corresponds to 100% relative humidity. Slight supersaturation ($RH > 1$) can occur, particularly before droplets have grown sufficiently to remove excess vapour, but large supersaturations are uncommon in the lower atmosphere.

It is sometimes said that "warm air can hold more water". A more precise statement is that the *saturation vapour pressure increases strongly with temperature.* Cooling moist air therefore increases its relative humidity even if the amount of water vapour has not changed, and sufficient cooling can bring the air to saturation.

4.2 Water-vapour mixing ratio

The water-vapour mass mixing ratio, r_v, is the mass of water vapour divided by the mass of dry air. From the ideal-gas laws,

$$r_v = \frac{\rho_v}{\rho_d} = \epsilon\frac{e}{P - e}, \qquad \epsilon = \frac{R_a}{R_v} \simeq 0.622. \tag{8}$$

When $e \ll P$, which is normally true in the atmosphere, $r_v \simeq \epsilon e/P$ is an excellent approximation.

5 Particles in the atmosphere

Atmospheric aerosol particles span a wide range of sizes and chemical compositions. Many contain soluble material such as ammonium sulphate, $(NH_4)_2SO_4$, sea salt or nitrate. Hygroscopic particles take up water as relative humidity increases, so their *wet* diameter can be substantially larger than their dry diameter. This matters for both visibility and aerosol mass.

A simplified Köhler relation for a solution droplet can be written

$$RH = \frac{n_w}{n_w + \nu n_s} \exp\left(\frac{4 M_w \sigma}{R_{\mathrm{gas}} T \rho_w D}\right), \tag{9}$$

where n_w is the number of moles of water, n_s is the number of moles of solute, M_w is the molar mass of water, σ is the surface tension of water, ρ_w is liquid-water density, D is the wet particle diameter and ν is an effective dissociation (van't Hoff) factor.

The first factor in Equation 9 represents the lowering of equilibrium vapour pressure by dissolved solute; the exponential term represents the curvature (Kelvin) effect. Together they describe how the equilibrium humidity depends on particle composition and size.

Mass conservation gives

$$\frac{\pi}{6}D^3\rho_{\mathrm{sol}} = n_w M_w + n_s M_s, \tag{10}$$

where M_s is the molar mass of the dry solute and ρ_{sol} is the density of the wet solution. If the volumes of water and solute are treated as additive,

$$\frac{n_w M_w + n_s M_s}{\rho_{\mathrm{sol}}} = \frac{n_w M_w}{\rho_w} + \frac{n_s M_s}{\rho_s}, \tag{11}$$

where ρ_s is the density of the dry solute.

For the practical calculations, the following approximate properties are useful:

Compound	NaCl	H_2SO_4	organic acid	NH_4NO_3
ν	2	2.5	1	2
ρ_s (kg m^{-3})	2160	1840	1500	1725
M_s (kg mol^{-1})	58.44×10^{-3}	98×10^{-3}	200×10^{-3}	80×10^{-3}

6 Visibility

Particles reduce visibility by scattering and absorbing light. In the deliberately simplified treatment used here, a monodisperse population of spherical particles has extinction coefficient

$$\beta_{\mathrm{ext}} = Q_{\mathrm{ext}}\frac{\pi D^2}{4}N_p, \tag{12}$$

where N_p is particle number concentration and Q_{ext} is the extinction efficiency. For the exercises we use $Q_{\mathrm{ext}} \simeq 2$, so

$$\beta_{\mathrm{ext}} \simeq 2\frac{\pi D^2}{4}N_p. \tag{13}$$

Visual range, V_{vis}, is inversely related to extinction:

$$\beta_{\mathrm{ext}} = \frac{K}{V_{\mathrm{vis}}}. \tag{14}$$

For consistency with the practical we use $K = 1.9$. The value of this constant depends on the contrast threshold used to define “visible”, so other texts may quote a different value.

Because hygroscopic particles grow as humidity rises, increasing relative humidity can increase D, raise β_{ext} and reduce visual range even when the dry aerosol amount has not changed.

7 Particulate matter

Particulate-matter concentration is usually reported as mass per unit volume of air, commonly in μg m^{-3}. PM_{10} and $PM_{2.5}$ refer to particles selected according to aerodynamic diameter. In this simplified exercise we treat the stated spherical particle diameter as the relevant size and assume each population is monodisperse.

For a spherical particle of diameter D and density ρ_p, the particle mass is

$$m_p = \frac{\pi}{6}\rho_p D^3, \tag{15}$$

so a population with number concentration N_p has mass concentration

$$C_m = N_p m_p = N_p\frac{\pi}{6}\rho_p D^3. \tag{16}$$

8 Questions to go through

8.1 Pressure

1. Calculate how many moles of air there are in a container of 2 m^3 at $P = 1000$ hPa and $T = 290$ K.
2. How many molecules is that? Use Avogadro's constant, $N_A = 6.022 \times 10^{23}$ mol^{-1}.
3. What is the density of air at $P = 1000$ hPa and $T = 280$ K?
4. What is the density of air at $P = 100$ hPa and $T = 210$ K?

8.2 Lapse rate

For these questions use 10 K km^{-1} for an unsaturated parcel and 6 K km^{-1} for a saturated parcel.

1. A dry thermal starts at the ground at $T = 290$ K and rises 3000 m. What is its final temperature? Assume no mixing.
2. A saturated thermal starts at 1000 m at $T = 280$ K and rises a further 200 m. What is its final temperature? Assume no mixing.
3. A thermal starts at the ground at $T = 300$ K. It rises 800 m before becoming saturated, then rises a further 300 m. What is its final temperature? Assume no mixing.

8.3 Humidity, partial pressure and mixing ratio

1. An air mass contains 3 g of water vapour per cubic metre of air at $T = 280$ K. What is the water-vapour partial pressure?
2. For the same air, calculate the saturation vapour pressure and hence the relative humidity. Is the air saturated?
3. The total pressure is 800 hPa. What is the water-vapour mass mixing ratio?
4. What is the approximate density of the air?

8.4 Köhler equation

For these questions use an ammonium-sulphate dry-particle density $\rho_s = 1770$ kg m^{-3} and molar mass $M_s = 132 \times 10^{-3}$ kg mol^{-1}.

1. A spherical ammonium-sulphate particle has dry diameter $D = 150$ nm. How many moles of solute does it contain?
2. If the same particle contains ten times as many moles of water as moles of solute, what is the density of the resulting solution? Use Equation 11.
3. What is the wet diameter? Use Equation 10.
4. At $T = 290$ K, at what equilibrium relative humidity will this wet particle exist? Use $\nu = 2$ and Equation 9.
5. What is the water-vapour partial pressure at this humidity and temperature?
6. Evaluate $n_w/(n_w + \nu n_s)$ and compare it with the full relative humidity. What does the difference tell you about the curvature term for this particle?

8.5 Visibility

1. A haze contains spherical particles of wet diameter $1\,\mu$m at a total number concentration of 1000×10^6 m^{-3}. Using $Q_{\text{ext}} = 2$, what is β_{ext}?
2. What visual range corresponds to this haze using $K = 1.9$?
3. What would increasing relative humidity tend to do to the visibility? Explain using the appropriate equations.

8.6 Particulate matter

1. A haze contains particles of diameter $1\,\mu$m, density 1003 kg m^{-3} and number concentration 1000×10^6 m^{-3}. What is the total particulate-matter loading in μg m^{-3}?
2. A haze contains two particle populations:
   - $D = 5\,\mu$m, $\rho_p = 1001$ kg m^{-3}, $N_p = 100 \times 10^6$ m^{-3};
   - $D = 1\,\mu$m, $\rho_p = 1003$ kg m^{-3}, $N_p = 800 \times 10^6$ m^{-3}.

   Calculate the total, PM_{10} and $PM_{2.5}$ mass loadings in μg m^{-3} using the simplified size convention stated above.

9 Answers to questions

9.1 Pressure

1. From Equation 1,
$$n = \frac{PV}{R_{\text{gas}}T} = \frac{(1000 \times 100) \times 2}{8.314 \times 290} \simeq 82.96 \text{ mol.}$$
2.
$$N = nN_A = 82.96 \times 6.022 \times 10^{23} \simeq 4.996 \times 10^{25} \text{ molecules.}$$
3. From Equation 3,
$$\rho_a = \frac{P}{R_a T} = \frac{1000 \times 100}{287 \times 280} \simeq 1.24 \text{ kg m}^{-3}.$$
4.
$$\rho_a = \frac{100 \times 100}{287 \times 210} \simeq 0.166 \text{ kg m}^{-3}.$$

9.2 Lapse rate

1.
$$T_{\text{final}} = 290 - (10)(3.0) = 260 \text{ K.}$$
2.
$$T_{\text{final}} = 280 - (6)(0.2) = 278.8 \text{ K.}$$
3. The parcel cools dry adiabatically for 0.8 km and then at the representative saturated rate for 0.3 km:
$$T_{\text{final}} = 300 - (10)(0.8) - (6)(0.3) = 290.2 \text{ K.}$$

9.3 Humidity, partial pressure and mixing ratio

1. 3 g m^{-3} = 3×10^{-3} kg m^{-3}. Therefore
$$e = \rho_v R_v T = (3 \times 10^{-3})(461)(280) = 387.2 \text{ Pa.}$$
2. From Equation 6,
$$e_s(280\,\text{K}) \simeq 992.6 \text{ Pa},$$
so
$$RH = \frac{387.2}{992.6} = 0.390,$$
or about 39.0%. The air is therefore unsaturated.
3. With $P = 800$ hPa $= 8.00 \times 10^4$ Pa,
$$r_v = 0.622\frac{387.2}{80000 - 387.2} \simeq 3.03 \times 10^{-3} \text{ kg kg}^{-1} = 3.03 \text{ g kg}^{-1}.$$
Using the approximation $r_v \simeq 0.622e/P$ gives 3.01 g kg^{-1}.
4. To the accuracy required here,
$$\rho_a \simeq \frac{80000}{287 \times 280} \simeq 0.995 \text{ kg m}^{-3}.$$

9.4 Köhler equation

1. The dry-particle mass is
$$m_s = \frac{\pi}{6}\rho_s D^3 = \frac{\pi}{6}(1770)(150 \times 10^{-9})^3 \simeq 3.13 \times 10^{-18} \text{ kg.}$$
Thus
$$n_s = \frac{m_s}{M_s} \simeq \frac{3.13 \times 10^{-18}}{132 \times 10^{-3}} = 2.37 \times 10^{-17} \text{ mol.}$$
2. Here $n_w = 10n_s = 2.37 \times 10^{-16}$ mol. Rearranging Equation 11 gives
$$\rho_{\text{sol}} \simeq 1226 \text{ kg m}^{-3}.$$

3. From Equation 10,
$$D = \left[\frac{6(n_w M_w + n_s M_s)}{\pi \rho_{\rm sol}}\right]^{1/3} \simeq 2.26 \times 10^{-7} \text{ m} = 226 \text{ nm}.$$
4. Substitution into Equation 9 gives
$$RH \simeq 0.842,$$
or about 84.2%.
5. At 290 K, Equation 6 gives $e_s \simeq 1936$ Pa. Therefore
$$e = RH\, e_s \simeq 0.842 \times 1936 \simeq 1.63 \times 10^3 \text{ Pa}.$$
6. $$\frac{n_w}{n_w + 2n_s} = \frac{10}{10+2} = 0.833.$$
This is close to the full value, 0.842. For this particle the curvature term changes the equilibrium RH only modestly, although it is not exactly unity.

9.5 Visibility

1. $$\beta_{\rm ext} = 2\frac{\pi(1 \times 10^{-6})^2}{4}(1000 \times 10^6) \simeq 1.57 \times 10^{-3} \text{ m}^{-1}.$$
2. $$V_{\rm vis} = \frac{1.9}{1.57 \times 10^{-3}} \simeq 1.21 \times 10^3 \text{ m},$$
or about 1.2 km.
3. Higher relative humidity causes hygroscopic particles to take up water and grow. Since $\beta_{\rm ext} \propto D^2$ in this simplified treatment, extinction increases and visual range decreases.

9.6 Particulate matter

1. $$C_m = (1000 \times 10^6)\frac{\pi}{6}(1003)(1 \times 10^{-6})^3 = 5.25 \times 10^{-7} \text{ kg m}^{-3} \simeq 525\ \mu\text{g m}^{-3}.$$
2. For the 1 μm population,
$$C_{1\mu\rm m} \simeq 420\ \mu\text{g m}^{-3}.$$
For the 5 μm population,
$$C_{5\mu\rm m} = (100 \times 10^6)\frac{\pi}{6}(1001)(5 \times 10^{-6})^3 \simeq 6552\ \mu\text{g m}^{-3}.$$
Both populations are below 10 μm in this simplified exercise, so
$$PM_{10} = \text{total} \simeq 6552 + 420 = 6972\ \mu\text{g m}^{-3}.$$
Only the 1 μm population contributes to $\text{PM}_{2.5}$, so
$$PM_{2.5} \simeq 420\ \mu\text{g m}^{-3}.$$